

Understanding the Transition from Memorization to Generalization in Diffusion Models for Cosmology

Jiaming Pan¹ Nicholas Kern¹ Dragan Huterer¹

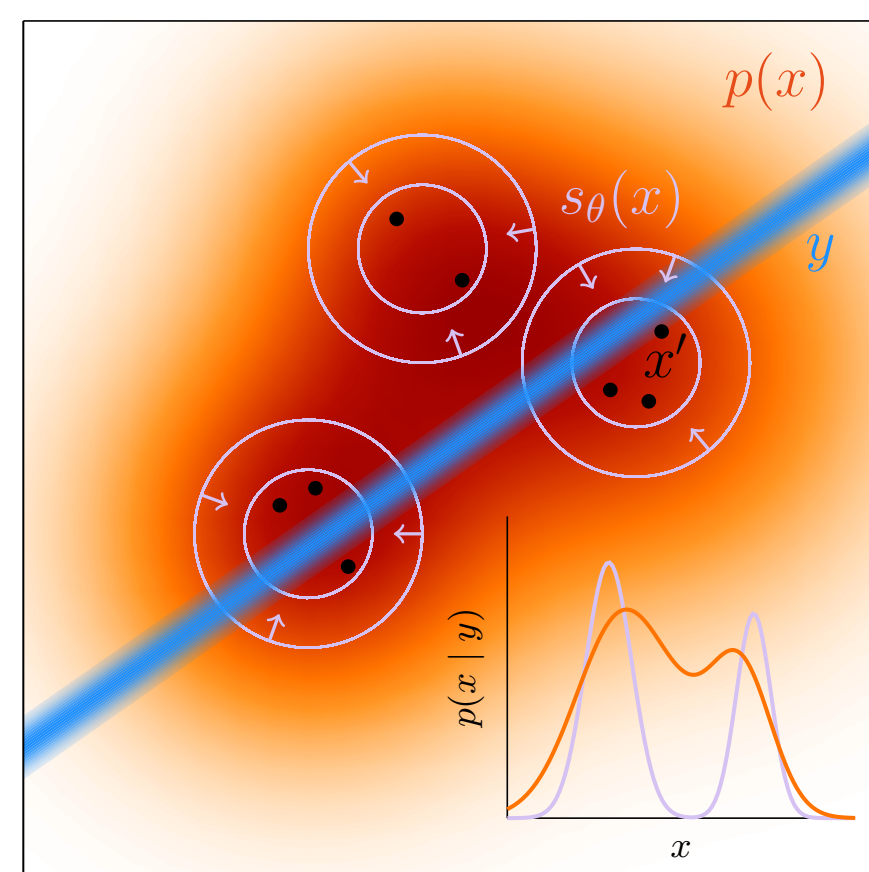
¹Department of Physics, University of Michigan • jiamingp@umich.edu • 2026 AstroAI Workshop, CfA

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The question

A trained diffusion model can produce mock HI maps in seconds instead of CPU-hours. That speed is only useful if the maps are genuinely new — a model that memorizes its training set gives a biased prior. We train diffusion models on CAMELS HI maps and ask:



1. When does the model stop **copying** training fields and start **sampling** the underlying distribution?
2. Does conditional generation actually **respond to the requested cosmology** θ ?
3. Are generated fields **statistically faithful** (one-point PDF, power spectrum)?

Toy picture: a memorized prior is a set of islands around the training data. The posterior gets pulled toward those islands even when the likelihood is sharp.

Data & models

- **CAMELS** (IllustrisTNG) LH set: 1000 simulations $\rightarrow$ 128 slices each $\rightarrow$ 128,000 candidate 128×128 HI maps.
- Six-parameter conditioning: $\theta = (\Omega_m, \sigma_8, A_{SN1}, A_{AGN1}, A_{SN2}, A_{AGN2})$.
- **Diffusion objective.** The model learns to undo noise: corrupt a real map, train a UNet to predict the velocity target at every noise level; the cosmology enters as conditioning.

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{x_0, \epsilon, t} \|f_\theta(x_t, t, \theta) - \text{target}\|_2^2.$$

Here the config uses `v_prediction: target = v_t = \sqrt{\alpha_t} \epsilon - \sqrt{1 - \alpha_t} x_0`. Ho et al. 2020 (DDPM).

- **Sampling:** 50-step DPM-Solver, $8\times$ faster than the 500-step DDPM baseline at matched quality.
- **Training-set sweep:** $N_{2D} = 2^6 \rightarrow 2^{15}$ fields, matched optimizer-update budget; runs on U-M Great Lakes HPC.

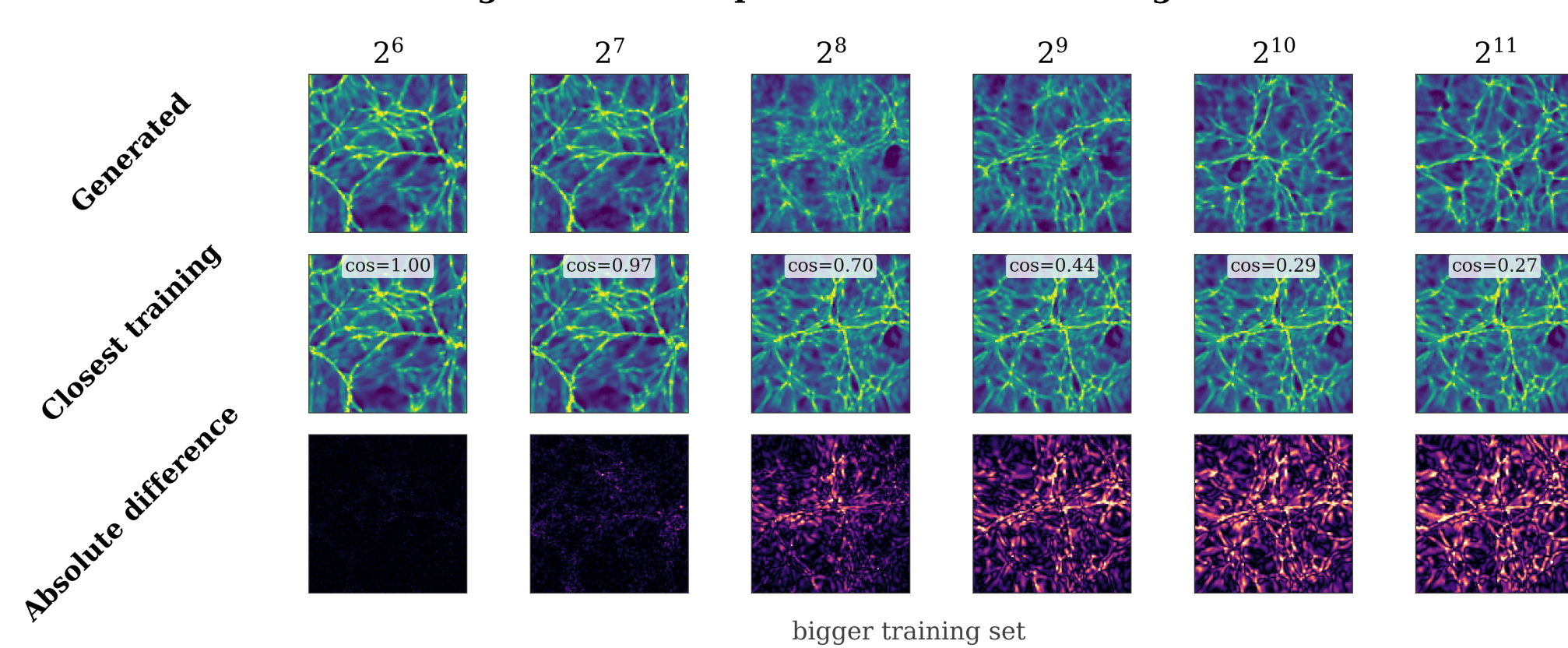
Detecting memorization

For each generated sample x_j , find its nearest training field in pixel, PCA, and SSCD embedding space:

$$s_j = \max_i \text{sim}(x_j, y_i), \quad \text{GL}(\tau) = 1 - \Pr[s_j > \tau]$$

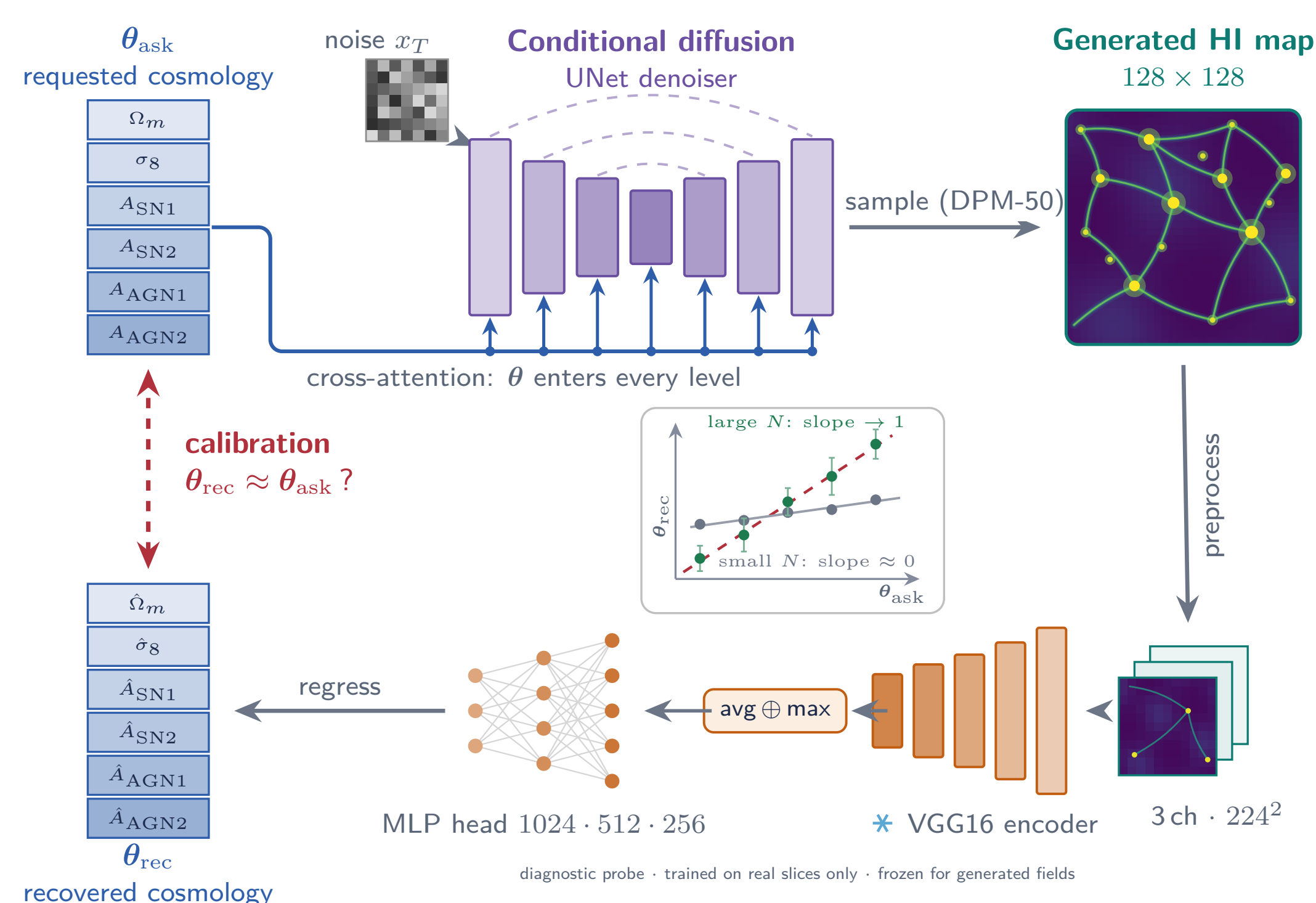
Here τ is set from train-vs-train nearest-neighbor similarity (95th percentile). Independent models should reproduce the same distribution, not the same training images.

UNet-64: generated samples and nearest training slices



Calibration: does the generator obey θ ?

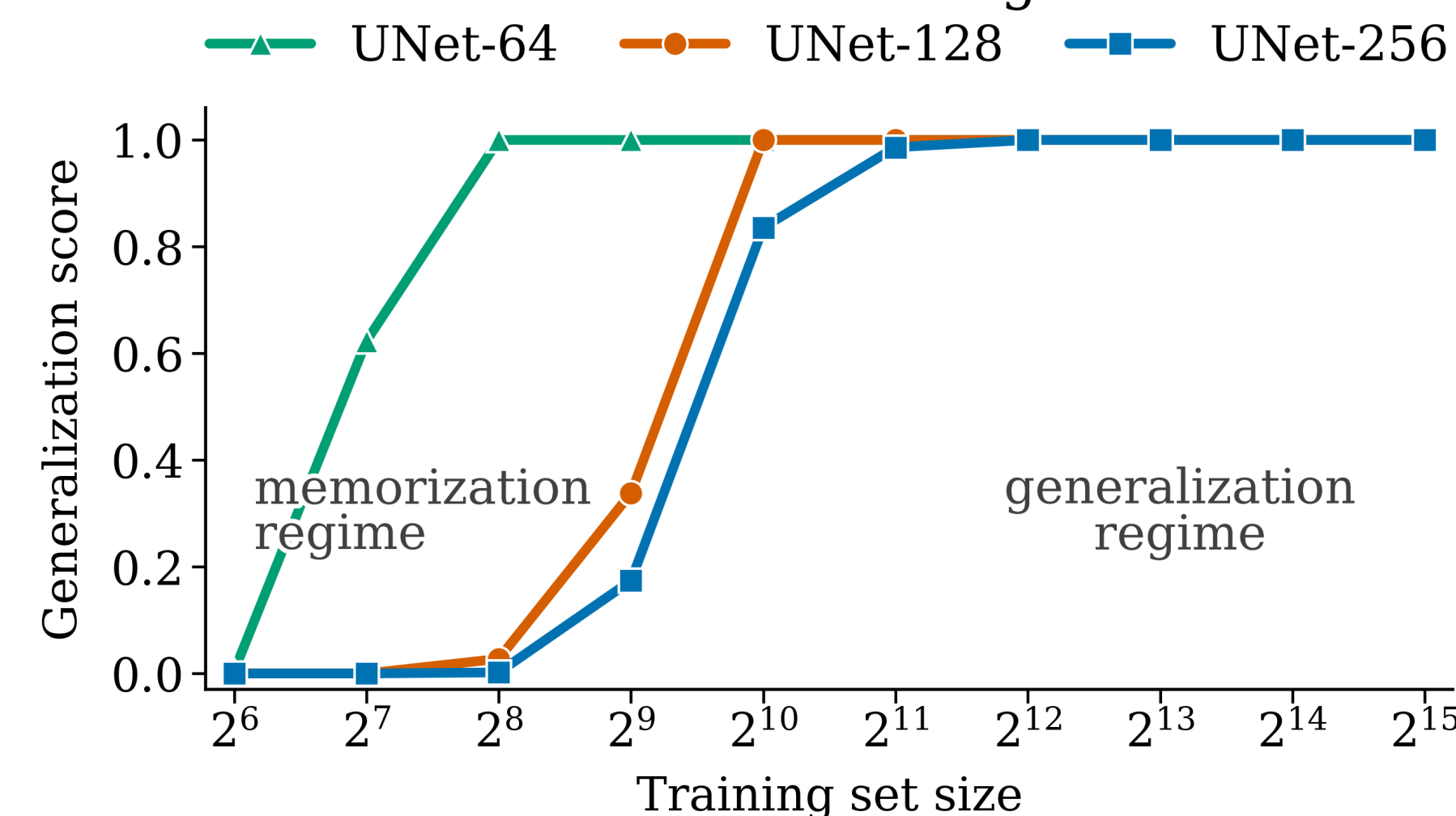
For a held-out sample, request maps at a chosen cosmology θ_{ask} . A **frozen encoder** then estimates θ_{rec} from the generated maps. If conditioning is working, θ_{rec} should move with the request, not stay near the training-set average.



Encoder setup: VGG16 is frozen (ImageNet); only the regression head is trained on *real* slices from non-held-out simulations. Held-out test simulations are used only for encoder validation and generator calibration.

Memorization $\rightarrow$ generalization transition

Generated fields become less training-set-like with more data

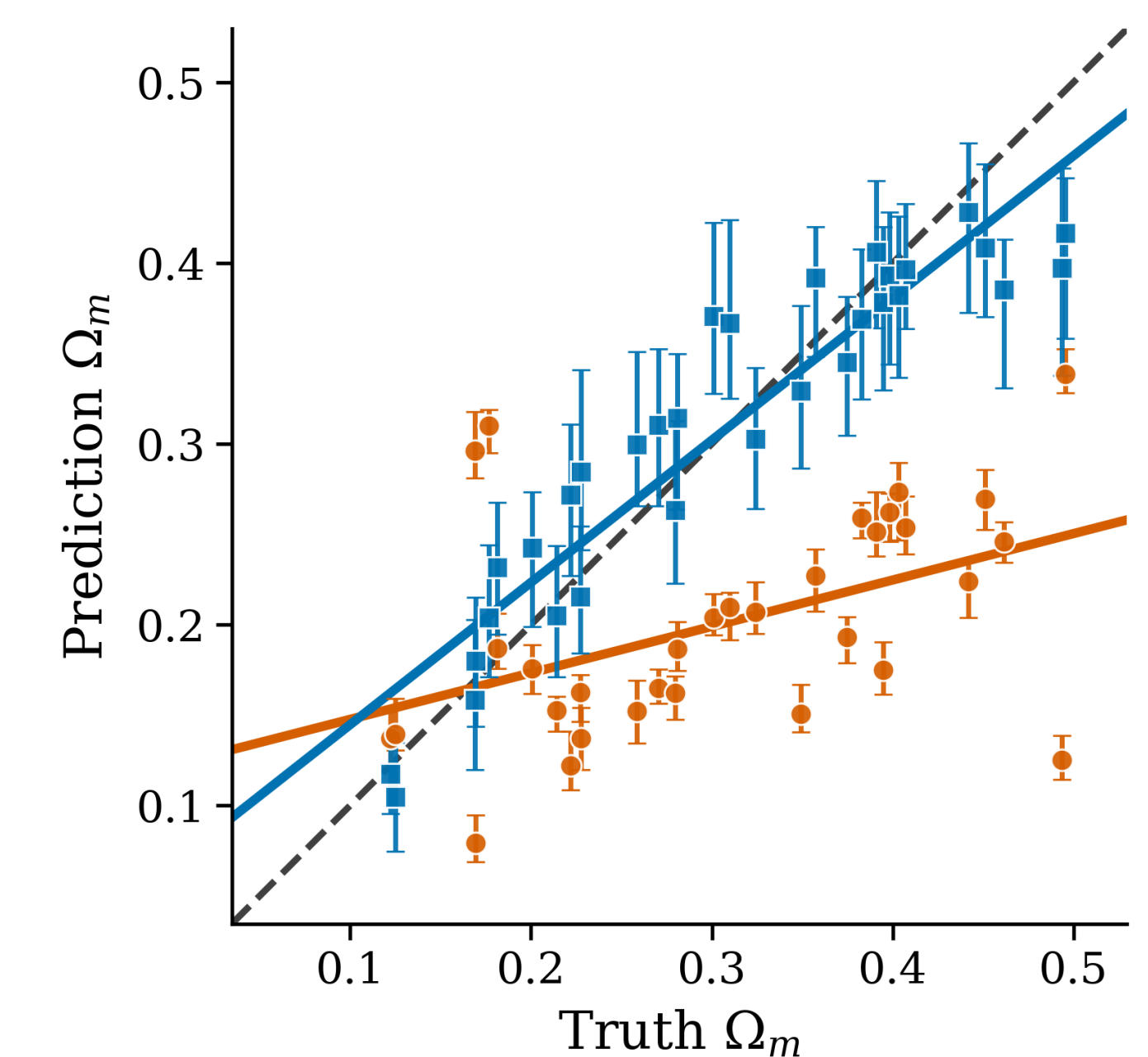


- Score uses a **q95-calibrated threshold** ($\tau = 95$ th pct train–train NN similarity): higher means generated fields are less training-set-like in PCA space.
- **Small N_{2D} :** generated maps remain close to training fields.
- **Larger N_{2D} :** UNet-64/128/256 move toward novel samples; the wider UNet-256 transition is shifted to larger data.

Calibration results

Generated HI fields recover Ω_m

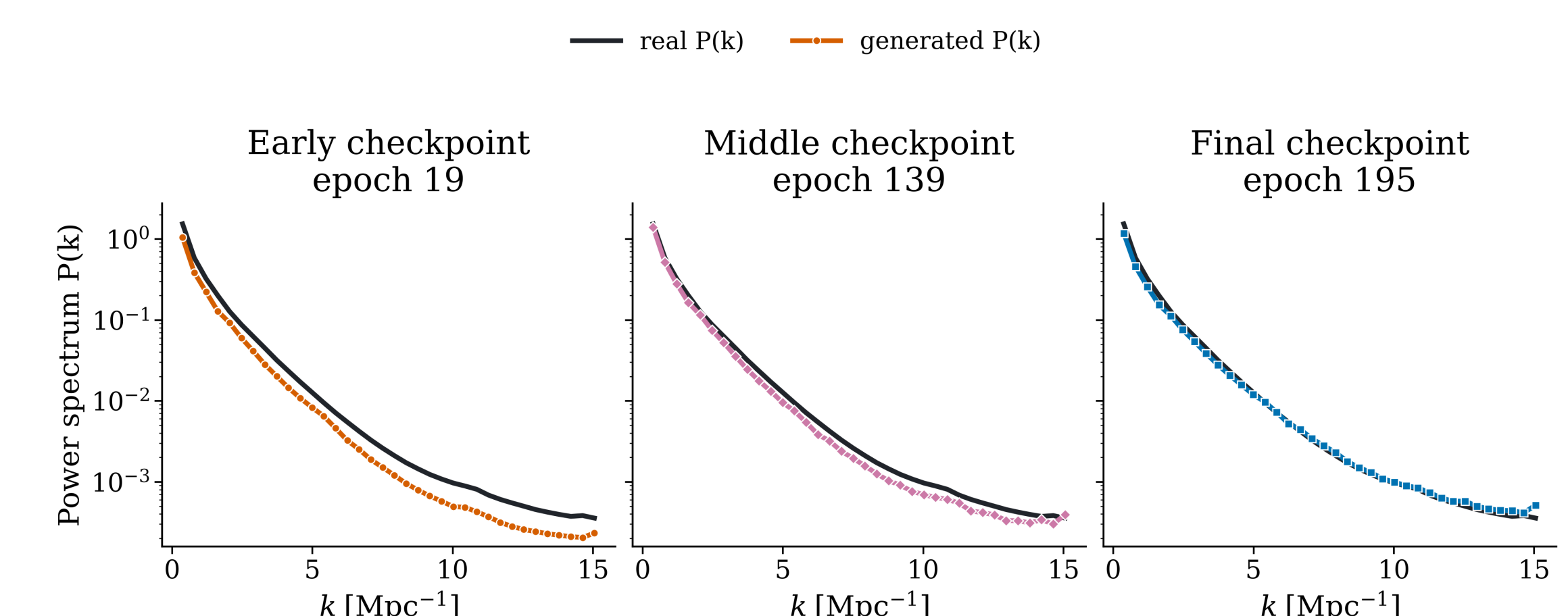
— ideal recovery ● mem. regime ($N_{2D} = 2^7$), slope=0.26 ■ gen. regime ($N_{2D} = 2^{14}$), slope=0.79



Main result: error bars show 16–84% seed scatter at fixed θ_{ask} . In the high-data regime, different noise seeds give a real spread and the intervals often bracket the requested Ω_m . Under memorization, many seeds collapse toward the same training-like fields: the bars can look tight while still being biased. This is why calibration matters more than just small scatter.

Physical fidelity

Generated power spectra approach the real reference during training



Across a single run, generated power spectra move toward the real reference from early to final checkpoints. Early checkpoints suppress power across much of k [Mpc⁻¹]; by the final checkpoint the generated $P(k)$ follows the real shape much more closely.

Takeaways

1. **Copying is measurable.** Nearest-neighbor and reproducibility tests expose when generated maps are training-set-like.
2. **More data changes behavior.** Increasing N_{2D} moves models from memorized slices toward new fields with realistic statistics.
3. **Calibration is the test.** Useful emulators must be novel, statistically faithful, and faithful to the requested cosmology.